

Use Case Specifications – Elemental Quest

Use Case 1: Start Game

Use Case: Start New Game

Primary Actor: Player

Goal: The player starts a new session of Elemental Quest.

Trigger: The player clicks the start button to start the game.

Preconditions:

1. The application is launched on the player's device.
2. The main menu is displayed.
3. Level data is available and successfully loaded.

Scenario:

1. The player opens the application.
2. The START menu is displayed on the screen.
3. The player clicks “Start Game” to begin the game.
4. The application transitions from the START menu to the game map.
5. The application loads the selected level, including:
 - Start point
 - Exit gate
 - Elemental Crystals
 - Gems
 - Shadow Sentinels
 - Hazards (lava and water)
6. The default character (Ember or Aqua) is placed at the start point.
7. The player's score is initialized to 0.
8. The game timer begins.

9. The application displays:
 - The map
 - Current score
 - Remaining time
 - Active character indicator
10. The player can now move the character using WASD keys and begin gameplay.

Postconditions:

1. The game is active.
2. The player can move, switch elements, and use abilities.
3. Enemies and hazards begin interacting with the player.

Use Case 2: Move Character

Use Case: Move Active Element

Primary Actor: Player

Goal: The player moves the currently active character across the map.

Preconditions:

1. The game is active.
2. The character (Ember or Aqua) is alive.

Scenario:

1. The player presses one of the movement keys (W, A, S, or D).
2. The application checks whether the intended direction is valid.
3. The application verifies that the player is not blocked by an object/ person.
4. If the move is valid, the character moves forward in that direction.
5. The application updates the character's position on the map.
6. Shadow Sentinels move closer to the player if necessary.
7. The screen refreshes to display updated positions.

Invalid Move:

1. If the target location contains a wall or barrier, the character does not move.
2. Enemies still move closer to the player to make the game more challenging.

Postconditions:

1. The character's position is updated or remains the same.
2. The game continues.

Use Case 3: Collect Elemental Crystals

Use Case: Collect Crystal

Primary Actor: Player

Goal: The player collects the crystals required to unlock the exit gate (complete the level)

Preconditions:

1. Either Ember or Aqua enters a space on the map that has a crystal.

Scenario:

1. The system detects that the crystal has been collected.
2. The crystal disappears from the map.
3. The score increases.
4. The crystal counter updates.

Postconditions:

1. The crystal is removed from the map.
2. Players' scores increase.

Use Case 4: Elemental Hazard Interaction

Use Case: Character Interacts with Hazard

Primary Actor: Player

Goal: The system processes the interaction with the hazard, and consequences arise for the player.

Preconditions:

1. A character enters a hazard space.

Scenario (Safe Interaction):

1. Ember enters lava → No penalty.
2. Aqua enters water → No penalty.

Harmful Interaction:

1. Ember enters water, or Aqua enters lava.
2. The system detects harmful interaction.
3. The character loses 1 of their life hearts.

Postconditions:

1. The level either continues or ends if that was your last life heart.

Use Case 5: Collision with Enemy

Use Case: Player Collides with Shadow Sentinel

Primary Actor: Enemy

Goal: Determine the outcome when the enemy reaches a character.

Preconditions:

1. The enemy and character occupy the same spot.

Scenario:

1. The system detects a collision.
2. The character loses 1 life heart, or the game ends.
3. A "Game Over" screen is displayed if that was their last life heart.

Postconditions:

1. The level ends if it was the last life heart.

Use Case 6: Level Completed

Use Case: Level Completion

Primary Actor: Player

Goal: Both characters safely reach their respective exit gates.

Preconditions:

1. All required crystals are collected.
2. Ember reaches the Fire Gate.
3. Aqua reaches the Water Gate.

Scenario:

1. Ember moves into the Fire Gate.
2. Aqua moves into the Water Gate.
3. The system verifies that all crystals were collected.
4. The timer stops.
5. Final score and time are displayed.
6. A "Level Complete" screen appears.

Only One Character Reaches Exit:

1. If only one character reaches their gate, the level does not complete.
2. The game continues until both reach their gates.

Postconditions:

1. Level is marked complete.
2. The player may proceed to the next level if they choose to still play.

Use Case 7: Level Completed

Use Case: Lose by Timeout

Primary Actor: System Timer

Goal: End the level when time runs out.

Preconditions:

1. Timer reaches zero.

Scenario:

1. Timer reaches zero.
2. The system stops all movement.
3. "Time Over" screen appears.
4. Final score is displayed.